

ADITYA ROY CHOWDHURY

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Education

Carleton University

MASc Electrical and Computer Engineering

Relevant Coursework: Uncertainty Evaluation in ML

Ottawa, Canada

Sep 2026 – May 2028

University of Southern California

MS Computer Science, GPA:3.87/4.00

Relevant Coursework: ML for Data Science, Advanced Computer Vision, Software Engineering

Los Angeles, USA

Aug 2024 – May 2026

Thapar Institute of Engineering Technology

BE Computer Science and Engineering, GPA:9.33/10.00

Relevant Coursework: Quantum Computing, Artificial Intelligence, Machine Learning

Punjab, India

Aug 2020 – June 2024

Technical Skills

Languages: Python, C++, MATLAB, Java

Machine Learning & CV: PyTorch, TensorFlow, CUDA

Backend & Databases: Flask, FastAPI, SQL

Tools & Cloud: Linux, Git, GCP, AWS, Docker

Experience

Research Assistant

Carleton University

Ottawa, Canada

Sep 2026 – Present

- Built and trained a 2025 Alzheimer's forecasting model (Neural Koopman Machine) from a paper with no released code, reproducing its reported accuracy (**r 0.535 vs 0.536**).
- Designed a module that unifies an embedding's different equivalent forms into one consistent representation defined only by intrinsic attributes and empirically verified its correctness.

Teaching Assistant

Carleton University

Ottawa, Canada

Sep 2026 – Present

- Assist faculty in delivering **SYSC 4806 (Software Engineering Laboratory)** to 130 students by leading a **30-student** lab section through Java/Spring Boot projects, assignments and technical debugging.

Machine Learning Engineer Intern

International Centre for Theoretical Physics

Trieste, Italy

Feb 2024 – Apr 2024

- Developed a **TensorFlow** object detection pipeline for adverse visibility conditions, utilizing knowledge distillation to compress **SpineNet 49S** backbone by **50%** by maximizing SSIM.
- Enabled FPGA deployment by porting **Python** model inference logic to **C++/HLS** with **Vivado**, generating hardware-synthesizable code.
- Collaborated with research scientists to produce **10K LOC** of Python/C++, building models and reusable tooling to streamline data processing and model evaluation.

Machine Learning Research Intern

Thapar Institute of Engineering Technology

Punjab, India

Jun 2023 – Jul 2023

- Led a **3-person** engineering team through Agile sprints to build a full-stack health monitoring system, coordinating integration of a **Flask** backend with a responsive **HTML/CSS** frontend for real-time visualization.
- Mitigated severe class imbalance in arrhythmia classification by constructing a tailored signal preprocessing pipeline, attaining **0.7 recall** with a **1D ResNet**.
- Ensured instant notification for critical cardiac events while minimizing false positives by engineering a hybrid alerting mechanism that combines ML predictions with static vital thresholds.

Machine Learning/Robotic Process Automation Intern

Indus Towers

Gurugram, India

Jun 2022 – Jul 2022

- Identified operational bottlenecks by analyzing 6 months of historical ITSM data across **10,000+ support tickets** using advanced pivot tables, resulting in accurately mapped volume patterns and quantified system downtime.
- Defined the scope for an outsourced email triage automation system by evaluating historical support workflows, resulting in a targeted plan to **automate 35%** of recurring Level-1 requests.
- Supported a senior manager in overseeing a vendor-built NLP project across **2 technical review cycles**, formulating technical queries for stakeholder meetings to ensure proposed solutions addressed internal constraints.